

The eye in systemic inflammatory diseases

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Systemic inflammatory diseases commonly affect the sclera, cornea, retina, and orbit, and can pose a serious threat to sight. They encompass both primary and secondary vasculitic disorders and specific granulomatous inflammatory conditions. As well as direct eye involvement from the systemic inflammatory process, there can be signs of ocular ischaemia due to carotid or ophthalmic arteritis, hypertensive retinopathy, and ocular complications such as chloroquine maculopathy related to anti-inflammatory drug treatment. Additionally, systemic infection relating to the eye, either as the result of primary infective disease processes or infection secondary to immunosuppression, might be mistaken as endogenous intraocular inflammation. Infection can closely mimic the ocular signs of endogenous inflammation, and in selected patients (such as those who have been immunosuppressed to treat vasculitis and who additionally have had invasive surgery, indwelling intravenous catheters, or systemic sepsis), it might be necessary to specifically exclude infection by the sampling and culturing of intraocular fluids and tissue.

We review common eye manifestations of systemic inflammatory disease because the pattern of eye involvement in such patients can provide valuable clues to the likely diagnosis or differential diagnosis. For example, in patients with scleritis and arthritis, rheumatoid arthritis is the likely associated systemic disease, whereas in patients with recurrent attacks of anterior uveitis and arthritis, seronegative B27 associated spondyloarthritis is most likely. In other patients, ocular signs such as retinal vasculitis can provide diagnostic features to make a diagnosis of Behçet's syndrome, or serous retinal detachments and panuveitis a diagnosis of Vogt-Koyanagi-Harada disease in patients with meningitis and poor vision. Understanding the range and patterns of ocular manifestations of systemic inflammatory disease is useful in the diagnosis and management of these patients.

Ocular features of systemic inflammatory disease

Episcleritis

Episcleritis is common, associated with minor ocular discomfort, and is typically evanescent and recurrent. Clinically, there is diffuse or localised oedema of the episclera with injection of episcleral blood vessels. The disorder is not vision threatening and does not need systemic treatment.

Scleritis

Scleritis is a serious and sight threatening ocular inflammatory disease, characterised by severe, often incapacitating, dull, boring, periocular pain that radiates to the face and scalp. Anterior scleritis is more common than posterior scleritis. Characteristically, the pain is worse at night, often waking the patient during the early hours. Patients are often investigated for alternative causes of severe headache and facial pain before scleritis is diagnosed. Rheumatoid arthritis accounts for about 25% of all patients with scleritis, Wegener's granulomatosis a further 15%, and 50% of patients have scleritis that is not associated with systemic disease.

In anterior scleritis, affected areas of sclera are injected, tender to palpation, and scleral nodules may develop. A

severe necrotising form of scleritis associated with occlusive vasculitis can result in frank scleral necrosis that may progress to perforation of the globe (figure 1).¹ This is accompanied by severe tenderness and pain. Clinical presentations of posterior scleritis depend on the location and extent of the posterior scleral involvement, and typical signs include: serous retinal detachment, optic disc swelling, painful acute vision loss, angle closure glaucoma, and choroidal detachment.

Keratitis

Keratitis commonly occurs with scleritis as a spillover of inflammation from the sclera and limbus to affect the peripheral cornea, resulting in corneal ulceration, opacification, or thinning, with occasional corneal perforation. Keratitis can also develop without scleritis and arise as inflammatory peripheral ulcerative keratitis (figure 2) or as corneal thinning either centrally or peripherally in apparently non-inflamed eyes.² Keratitis is treated similarly to scleritis but might also need corneal graft surgery.

Uveitis and retinal vasculitis

Uveitis refers to inflammation of the uveal tract and is classified anatomically. It has a wide range of inflammatory and infective causes, in addition to many recognisable ocular uveitic syndromes. Anterior uveitis is by far the most frequent pattern of disease, typically acute and recurrent (it can be associated with the HLA

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Search strategy and selection criteria

The MEDLINE database was searched by use of PubMed to identify articles containing "ocular", "eye", "uveitis", "scleritis", "keratitis", "retinal vasculitis systemic vasculitis", and the specific vasculitic diseases discussed in this Seminar. Additional references were obtained from bibliographies of papers retrieved using this strategy. As additional sources and cross checks for the reliability of the search strategy, we reviewed comprehensive textbooks.



Figure 1: Necrotising anterior scleritis

B27 spectrum of diseases), and usually responds readily to local corticosteroid treatment. Intermediate, posterior, and panuveitis are far less frequent than anterior uveitis; they are typically chronic, and more often associated with systemic inflammatory and infective processes. Panuveitis commonly represents a serious threat to vision and often needs systemic therapy.

Retinal vasculitis (figure 3) is a sign of posterior uveitis and can occur with a wide variety of ocular inflammatory and infective processes as well as in patients with systemic inflammatory disease. Vasculitis is sometimes the dominant clinical feature, and it can be diagnostically useful to separate such patients from the broader spectrum of those with uveitis. Clinical features are: sheathing of retinal vessels, retinal haemorrhages, macular and retinal oedema, optic disc swelling, and retinal vascular occlusion, all of which are variably associated with vitreous inflammation.

Retinal vasculitis most commonly affects the retinal veins and capillaries, and does not imply a specific diagnosis. Isolated retinal arterial vasculitis is uncommon, but typically is seen in patients with Behçet's syndrome and systemic lupus erythematosus. Fluorescein angiography is used to assess disease extent and severity. Treatment of retinal vasculitis depends on the cause and immediacy of the threat to vision, and commonly uses systemic corticosteroid therapy.

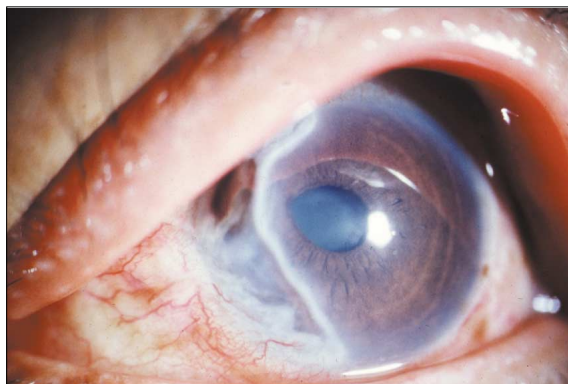


Figure 2: Peripheral ulcerative keratitis

Systemic vasculitis and the eye

The primary systemic vasculitides encompass a diverse range of pathological processes and organ involvement, with inflammation targeting the arteries, veins, and capillary beds (table 1;³ leucocytoclastic vasculitides have not been reviewed because they rarely affect the eye). The vasculitic process can target specific organs such as the kidneys, respiratory tract, and sinuses (eg, in Wegener's granulomatosis); whereas in other disorders, it targets specific tissues distributed throughout the body (eg, cartilage in relapsing polychondritis). Pretreatment tissue diagnosis should be obtained if possible, and urine analysis remains the most important investigation because prognosis is often determined by the extent of renal involvement. Raised erythrocyte sedimentation rates (ESR) and C-reactive protein concentrations (CRP) are non-specific markers of inflammation, whereas antineutrophil cytoplasmic antibodies (ANCA) are frequently associated with primary vasculitis (but are not diagnostic). Table 2 summarises the ocular manifestations seen commonly in patients with vasculitis.

Giant cell arteritis

Other terms for giant cell arteritis include temporal or cranial arteritis. This disease is a necrotising vasculitis targeting large and medium-sized arteries and mainly affects the white population with a female predominance. Vascular inflammation has a preference for the aorta and extracranial arteries. Although the disease is rare in individuals aged less than 50 years, the likelihood of diagnosis increases continuously with age thereafter. Giant cell arteritis is about 20 times more common in people aged more than 90 years compared with those aged 50–60 years.⁴ Giant cell arteritis forms a clinical spectrum with polymyalgia rheumatica, which is characterised by arthralgia and myalgia of the shoulder and pelvic girdle (but not associated with ocular changes). Increased ESR and CRP can help diagnose both disorders, and giant cell arteritis can be confirmed by a positive result on a temporal artery biopsy.

Visual loss from giant cell arteritis is an ocular emergency and a history of headaches, scalp tenderness, and jaw claudication demands urgent therapy. This treatment minimises the risk of total visual loss resulting from ischaemic optic neuropathy, which is secondary to vasculitis affecting end arteries (that supply the optic nerve head) or central retinal artery occlusion (from involvement of the ophthalmic artery). Ocular involvement is a fundamental feature of giant cell arteritis and occurs in more than 80% of patients. It is commonly followed within hours or days by similar severe visual loss in the second eye, unless treated with high-dose corticosteroids immediately. Established visual loss is rarely reversible.⁵

Corticosteroid treatment should be started immediately once the diagnosis is considered likely on clinical grounds, and a temporal artery biopsy

undertaken within 3–4 days. The histopathologist should also be informed that corticosteroid treatment has commenced. Because of the focal nature of the vasculitis, up to 15% of biopsies may be negative, and biopsies of both superficial temporal arteries might be needed to confirm the diagnosis.⁶ A raised ESR is also useful for diagnosis, but it could be normal. A normal ESR does not exclude giant cell arteritis.

Giant cell arteritis responds well and quickly to corticosteroids (40–60 mg per day), and such treatment prevents arteritic complications.^{7,8} Many ophthalmologists would use 80–100 mg prednisolone as a starting dose in patients presenting with visual symptoms and give at least one initial dose of 500–1000 mg IV methylprednisolone.⁹ The time before the initiation of corticosteroid treatment is the most important factor in outcome prediction in patients presenting with symptoms of visual impairment. In those who were treated within 24 h after loss of vision, improvement was achieved in more than 50%.¹⁰

Little evidence-based information exists on the reduction rate of steroid dose after initial symptoms are controlled and weekly decrements (of ≤ 5 mg) have been proposed down to 10 mg per day. Giant cell arteritis has a remarkable tendency to relapse during the tapering of steroids and the addition of weekly oral methotrexate has shown improvement of symptoms.¹¹ The newer biological drugs, especially those that block TNF α , might be beneficial in problematic patients.¹²

Takayasu's arteritis

Takayasu's arteritis is an uncommon disorder that occurs worldwide. It most typically presents in women of childbearing age and diagnosis is often delayed. Vascular occlusion leads to claudication, headaches, dizziness, syncope, visual changes, dyspnoea, palpitations, and even carotid artery tenderness, and examination often reveals absent pulses, bruits, and asymmetric blood pressure. Non-specific features include fever, night sweats, malaise, weight loss, arthralgia, myalgia, and mild anaemia.¹³

The granulomatous vasculitis affects large vessels, predominantly the aorta, main aortic branches, and pulmonary arteries, leading to the thickening of the vessel wall, fibrosis, stenosis, and thrombus formation. A more acute inflammatory process can actually destroy the arterial media with aneurysm formation. Hypertension is an important contributor to both morbidity and mortality. No good serological markers are known for this ANCA-negative disease, and diagnosis is made on the basis of clinical features and supported by angiographic evidence, after other causes of large vessel abnormalities have been excluded.

Ocular abnormalities arise in 40% of patients and include: microaneurysms, haemorrhages, neovascularisation of the retina, glaucoma, hypotony, and rubeosis iridis; they occur as a consequence of hypertension or ocular ischaemia from carotid occlusion, rather than direct ocular involvement by the vasculitis.^{13–15}



Figure 3: Diffuse retinal vasculitis

Corticosteroids remain the mainstay of treatment for Takayasu's arteritis, and approximately half of patients will respond to steroids;¹⁶ however, no controlled studies have been undertaken. Azathioprine has retrospectively been shown to ameliorate systemic symptoms and halt progression of angiographic lesions.¹⁷ Surgical treatment of critical vascular stenoses might be needed.

	Primary	Secondary
Large arteries	Giant cell arteritis Takayasu's arteritis	Aortitis associated with rheumatoid arthritis Infection (eg, syphilis and tuberculosis)
Medium arteries	Classic polyarthritis nodosa Kawasaki disease	Hepatitis-B-associated polyarthritis nodosa
Small vessels and medium arteries	Wegener's granulomatosis* Churg-Strauss syndrome* Microscopic polyangiitis*	Vasculitis secondary to rheumatoid arthritis, systemic lupus erythematosus, and Sjögren's syndrome Drugs Infection (eg, HIV)
Small vessels (leucocytoclastic)	Henoch-Schönlein purpura Cryoglobulinaemia Cutaneous leucocytoclastic angiitis	Drugs† Hepatitis-C-associated infection

*Diseases most commonly associated with ANCA. †Such as sulphonamides, penicillin, and thiazides. Adapted from reference 3.

Table 1: Classification of systemic vasculitis according to targeted vessel

	Clinical features	Risk of vision loss	Systemic disease
Scleritis	Pain, scleral injection and tenderness, scleral nodules, scleral necrosis	Moderate	Rheumatoid arthritis, Wegener's granulomatosis, relapsing polychondritis
Keratitis	Ulceration, thinning, perforation	Moderate	Rheumatoid arthritis, Wegener's granulomatosis
Uveitis	Anterior chamber cells Vitreous cells Choroiditis, retinitis, macular oedema, optic disc swelling	Moderate	Sarcoidosis, Behçet's syndrome, Vogt-Koyanagi-Harada disease
Retinal vasculitis*	Retinal haemorrhages, cotton wool spots, occluded vessels	High	Systemic lupus erythematosus, microscopic polyangiitis, polyarteritis nodosa, Wegener's granulomatosis
Orbital Inflammation	Proptosis, palpable mass, vision loss	Low	Polyarteritis nodosa, Wegener's granulomatosis
Optic neuropathy	Vision loss	High	Giant cell arteritis, Churg-Strauss syndrome

*In fact, retinal vasculitis is merely a clinical sign of posterior uveitis, but because it is the predominant sign in some patients, many clinicians separate patients with this clinical feature into a separate diagnostic category.

Table 2: Ocular manifestations of systemic vasculitis

Polyarteritis nodosa

Polyarteritis nodosa is rare, occurs most frequently in older men, and affects medium-sized or small muscular arteries. The disease results in weight loss, testicular pain, myalgia, hypertension, and mononeuropathy or polyneuropathy, and is typically ANCA-negative. It might also occur in association with hepatitis B infection, HIV infection, rheumatoid arthritis, Sjögren's syndrome, and myelodysplasia.

Ocular involvement in polyarteritis nodosa is uncommon and consists of scleritis, peripheral ulcerative keratitis, or orbital inflammation that usually presents as proptosis.¹⁸ Ocular involvement in polyarteritis nodosa is treated on its merits and depends on the severity of the inflammation and threat to vision. Use of cytotoxic drugs and steroids are vital to the treatment of this disease.

Kawasaki disease

Kawasaki disease is an acute vasculitis of early childhood that especially affects the coronary arteries. This condition has now surpassed rheumatic fever as the most common cause of acquired heart disease in children in developed countries.¹⁹ The disease is characterised by fever, rash, conjunctival injection, brawny induration of hands and feet with erythema (confined to the palms and soles), and lymphadenopathy. Coronary artery aneurysms and ectasia might be sequelae of acute vasculitis in up to 25% of patients. An infectious pathogen has been suggested by epidemiological studies, although it remains unidentified.²⁰

Conjunctival injection is a sign of Kawasaki disease and is due to vasculitis of the conjunctival vessels. A small number of patients also develop anterior uveitis. The uveitis and conjunctivitis usually resolve with topical corticosteroid treatment. Intravenous immunoglobulin therapy in conjunction with high-dose aspirin

for systemic manifestations reduces the incidence of coronary artery abnormalities.^{21,22}

Wegener's granulomatosis

Wegener's granulomatosis is a small-vessel vasculitis with a predilection for the upper respiratory tract, lungs, and kidneys. In individuals with severe widespread disease, typical laboratory findings include anaemia, neutrophil leucocytosis, and positive cytoplasmic ANCA (cANCA) and proteinase 3 (PR3; a marker for the disease) results. A limited form of Wegener's granulomatosis is confined to the eyes, orbits, and sinuses, and although it is considered to have an improved prognosis, it can be very destructive locally, leading to substantial morbidity. Sinus involvement is very common and is an important biopsy source for confirmatory histological diagnosis.

Ophthalmic involvement occurs in about 40% of patients and can be the clinical presentation of the illness. Orbital involvement manifests as proptosis, painful extraocular muscle weakness, or orbital mass lesions.²³ Wegener's granulomatosis is the second most common systemic disorder associated with scleritis and accounts for up to 15% of cases.²⁴ Both necrotising and non-necrotising scleritis can be seen, typically associated with peripheral keratitis. Retinal vasculitis is the most characteristic pattern of intraocular involvement, but rarely arises without other clinical features of the disease. A wide range of other ocular manifestations has been reported.²⁵

Induction therapy for systemic disease usually is high-dose corticosteroids and cyclophosphamide for 3–6 months, followed by azathioprine or methotrexate in the long-term to maintain remission. Disease confined to the nose and orbits can produce substantial morbidity, and therefore should be assessed regularly and treatment instigated if evidence of local progression is seen. Ocular and orbital involvement usually need therapy.

Churg-Strauss syndrome

Churg-Strauss syndrome is a rare small-vessel vasculitis characterised by asthma, eosinophilia, and nasal symptoms. Other manifestations include peripheral neuropathy, CNS involvement, vasculitic purpura, livedo reticularis, gut involvement, cardiomyopathy, arthralgia, and myalgia. Investigations reveal presence of perinuclear ANCA/myeloperoxidase (pANCA/MPO; a valuable marker for many inflammatory disorders) in 70% of affected individuals, an eosinophilia count of more than 1500 cells per mL,³ and an increase in serum IgE concentrations. Scleritis, orbital inflammation, and ischaemic optic neuropathy are typical ocular features of Churg-Strauss syndrome. The syndrome usually responds well to corticosteroids, although additional immunosuppressive drugs (such as cyclophosphamide) might be necessary.

Microscopic polyangiitis

Microscopic polyangiitis is used to distinguish this small-vessel vasculitis from classic polyarteritis nodosa (which affects medium-sized arteries). It is typically associated with rapidly progressive glomerulonephritis, whereas the renal damage in polyarteritis nodosa is usually secondary to hypertension. Clinical features of microscopic polyangiitis include malaise, weight loss, myalgia and arthralgia, mononeuritis multiplex, and skin involvement. Rashes are present in at least two-thirds of patients, specifically purpura, livedo reticularis, and urticaria. Renal and pulmonary involvement can be life threatening. ANCA are detected in about 80% of affected patients (typically as the pANCA/MPO-positive pattern).

Scleritis and retinal vasculitis are the most frequent ocular manifestations of microscopic polyangiitis. Corticosteroid and cyclophosphamide treatment is used initially to control microscopic polyangiitis, switching to azathioprine or methotrexate once the vasculitis is in remission.

Rheumatoid arthritis

Rheumatoid arthritis is the most common chronic inflammatory disease of the joints. Although its major manifestations relate to the musculoskeletal system, it can affect any organ or system in the body; hence the term rheumatoid disease is probably more appropriate. It is a symmetrical erosive polyarthritis that affects more women than men. Laboratory findings usually include a raised ESR and CRP, anaemia, thrombocytosis, and leucocytosis. Rheumatoid factors in high titre are often present in patients with systemic complications and ANCAs are seen in those with rheumatoid vasculitis, but these individuals are often not PR3 or MPO-positive.

The most common ocular problem in rheumatoid arthritis is dry eye from secondary Sjögren's syndrome. Dry eyes cause many symptoms but rarely threaten vision and usually respond to ocular lubricants. Episcleritis is also common, benign in nature, and responds readily to topical non-steroidal anti-inflammatory drugs or topical corticosteroids.

Scleritis is usually seen in patients with advanced disease and other extra-articular manifestations such as digital arteritis, cutaneous ulceration, palpable purpura, peripheral neuropathy, and arteritis of the viscera and CNS. Rheumatoid scleritis is characterised by a granulomatous vasculitis affecting the vascular plexuses of the sclera and episclera with mononuclear cell infiltration and fibrinoid necrosis of blood vessel walls.¹⁸ Onset of vasculitis in rheumatoid arthritis is potentially life threatening, and in one study,²⁶ the development of ocular manifestations was associated with a 50% mortality rate over 5 years without systemic immunosuppressive therapy.

Rheumatoid-associated keratitis is common in patients with active scleritis. It might also develop

without scleritis, and present as inflammatory peripheral ulcerative keratitis (figure 2) or as corneal thinning either centrally or peripherally in non-inflamed eyes.²⁶ Scleromalacia perforans is a very rare type of scleritis that is due to obliterative endarteritis of the scleral vessels. It is seen exclusively in patients with longstanding rheumatoid arthritis, is clinically silent and pain-free, and is becoming extremely rare with improved treatments for rheumatoid arthritis.²⁷

Scleritis does not respond to local therapy and needs systemic treatment with oral non-steroidal anti-inflammatory drugs, corticosteroids, and additional immunosuppressive drugs, according to the severity of inflammation and the threat to vision. Rheumatoid eye disease is often sight threatening. Anti-TNF α drugs have great potential in the treatment of sight-threatening ocular vasculitis, but studies of their use in ocular disease have not been sufficient to define their role.²⁸

Sjögren's syndrome

Sjögren's syndrome typically affects women aged 40–50 years. It results from an autoimmune lymphocytic infiltration of the exocrine glands, producing predominantly dry eyes and mouth, often with major salivary gland enlargement. Individuals with primary Sjögren's syndrome have non-erosive arthritis, cutaneous vasculitis, peripheral neuropathy, fatigue, depression, drug allergies, and food sensitivities. Secondary Sjögren's syndrome presents as dry eyes and mouth in patients with rheumatoid arthritis, systemic lupus erythematosus, and (less commonly) other autoimmune diseases. Abnormal investigations typical of primary Sjögren's syndrome include hypergammaglobulinaemia and the presence of anti-Ro (SS-A) and anti-La (SS-B) anti-nuclear antibodies.

Gritty, burning, irritable, dry eyes that increase in severity throughout the day and are exacerbated by wind, air-conditioning, and other drying environments, are classic symptoms of the aqueous tear deficiency that characterises Sjögren's syndrome. Clinical features include deficient tear film with increased mucous production, reduced Schirmer tests, and positive ocular surface staining with vital dyes such as Bengal rose, lissamine green, and fluorescein. Clinical findings and Schirmer tests vary a great deal over time and in different patients. Microbial keratitis is very uncommon in patients with Sjögren's syndrome. Occasionally, patients might develop corneal melting that threatens the integrity of the globe.

The ocular and oral manifestations of Sjögren's syndrome cause major impairment to quality of life and treatments so far are essentially palliative with tear supplements, oral saliva substitutes, and lacrimal canaliculal duct occlusion. Some patients have improved salivary and lacrimal gland secretion if treated with secretagogues such as bromhexine. The immune-mediated ocular surface inflammation and destruction of

secondary ocular surface lacrimal glands seen in Sjögren's syndrome might improve with topical cyclosporin therapy, and there are substantial improvements in patient symptoms and quality of life associated with such treatment.²⁹ This therapy is available commercially in the USA and locally manufactured forms have been used for many years in other countries.

Systemic lupus erythematosus

Systemic lupus erythematosus is a multisystem autoimmune disease, characterised by skin rashes, photosensitivity, oral ulceration, pleurisy, and renal and neuropsychiatric involvement. Tests commonly show leucopenia, lymphopenia, thrombocytopenia, positive antinuclear antibody results, and in 50% anti-double-stranded DNA antibodies as well.

Systemic lupus erythematosus is a common cause of secondary Sjögren's syndrome. Mild and frequently recurrent episcleritis might develop in affected patients and needs topical non-steroidal anti-inflammatory drugs or an occasional short course of topical corticosteroids. Scleritis could also develop, which is usually non-necrotising and responds well to oral non-steroidal anti-inflammatory drugs or oral steroids. Retinal vasculitis is a common ocular manifestation of systemic lupus erythematosus. It is an arteriolar vasculitis that produces multiple retinal cotton wool spots and occasionally, larger retinal arterial occlusions. Retinal venulitis might also develop. Both could result in substantial visual loss. Retinal vasculitis must be distinguished from the hypertensive vascular changes that might arise in systemic lupus erythematosus with renal involvement.

Treatment of systemic lupus erythematosus depends on the severity of the disease, with corticosteroids being very useful in acute management, but disease-modifying drugs such as cyclophosphamide, azathioprine, ciclosporin, and tacrolimus could also be considered if vision-threatening eye involvement occurs.³⁰

Behçet's syndrome

Behçet's syndrome is a systemic vasculitis of unknown cause that affects veins and arteries of all sizes, producing recurrent mucocutaneous lesions and frequent ocular involvement. Musculoskeletal, neurological, and gastrointestinal manifestations are also seen. Prevalence of the disease is highest in Turkey, the Middle East, Japan, and Korea. Age of onset is most commonly 20–30 years in men, and young patients have a more severe disease course. Behçet's syndrome has been associated with the HLA-B51 phenotype, but strength of this association varies worldwide. Although diagnosis is clinical, criteria such as those of the International Study Group³¹ (panel) are useful. Neutrophilia and mild increases in ESR and CRP might be seen, but these factors do not strongly correlate with disease activity.

Ocular involvement commonly leads to pronounced morbidity as episodes of severe panuveitis are associated

Panel: Diagnostic criteria for Behçet's syndrome from the International Study Group³¹

Recurrent oral ulceration (aphthous or herpetiform) recurring more than three times in 12 months, and two of the following (in the absence of other clinical explanations):

- recurrent genital ulceration
- eye lesions: anterior uveitis, posterior uveitis, cells in the vitreous by slit-lamp examination, or retinal vasculitis
- skin lesions: erythema nodosum, pseudofolliculitis, papulopustular lesions, or acneiform nodules in postadolescent patients not on corticosteroid treatment
- pathergy test (skin hyper-reactivity to simple trauma) at 24–48 h

with occlusive vasculitis affecting both the retinal arteries and veins, which result in progressive, irreversible, ischaemic damage to the retina and optic nerve.³² Anterior uveitis can be severe enough to induce hypopyon formation. Visual loss might also develop as a result of vasculitis affecting the posterior visual pathway, or from severe papilloedema (due to raised intracranial pressure from intracranial venous sinus thrombosis).³³ Cataract and glaucoma are frequent ocular complications of the uveitis. Progressive loss of vision and blindness from severe occlusive vasculitis and optic atrophy is common despite aggressive immunosuppressive therapy,³³ and represents the major morbidity of Behçet's syndrome in many patients.

Some patients develop isolated mild anterior uveitis that can be treated adequately with local corticosteroids. Typically, patients have episodes of severe uveitis and retinal vasculitis that progressively damage vision. As a result, they need high-dose oral and intravenous corticosteroids to control acute exacerbations of uveitis and continuous systemic immunosuppression (eg, ciclosporin and azathioprine) to delay onset of severe visual loss and minimise long-term corticosteroid use.^{34–36} Cyclophosphamide and chlorambucil have been used empirically in the past, but the risk of neoplasia limits their use. The beneficial effect of anti-TNF α therapy has been reported in a few small studies and controlled trial results are now awaited.³⁷

Relapsing polychondritis

Relapsing polychondritis is a rare, chronic, multisystem disorder characterised by recurrent, episodic inflammation of cartilaginous structures that results in tissue destruction. The vasculitic process affects small, medium, and large vessels in the ears, eyes, nose, laryngobronchial and costal cartilages, and joints. Constitutional symptoms are common and there could be involvement of the heart, blood vessels, and inner ear. No specific serological tests are known and diagnosis is based on characteristic clinical manifestations.

Ocular involvement in relapsing polychondritis occurs in 50% of individuals and typically presents as non-necrotising recurrent scleritis.³⁸ Ocular involvement often culminates with diagnosis of the underlying condition, because there is severe pain that demands medical consultation, whereas symptoms of cartilage (nasal and auricular) inflammation might be easily dismissed by the patient. Other ocular involvement is rare.

Corticosteroids are the mainstay of treatment during acute attacks, and cytotoxic drugs such as cyclophosphamide, azathioprine, and methotrexate have been used as additional therapy. No controlled studies of these drugs have been undertaken.

Systemic inflammatory disease

Endogenous inflammatory disorders that frequently target the eye include granulomatous diseases, such as sarcoidosis with its protean clinical manifestations and frequent involvement of the eye, and other rare disorders, such as Vogt-Koyanagi-Harada disease (in which ocular involvement is a cardinal feature with the eye representing the major morbidity from the inflammatory process). Confirmation of the diagnosis is often difficult in these disorders and usually relies heavily on clinical signs rather than histological diagnosis.

Sarcoidosis

Sarcoidosis is a common granulomatous inflammatory disorder of unknown cause that typically affects the respiratory system but can also damage the skin, eyes, neurological system, and liver.³⁹ Only histological examination of affected tissue can confirm clinical diagnosis, although radiological, pulmonary function tests and bronchoalveolar lavage fluid have characteristic abnormalities.³⁹ Raised serum angiotensin-converting-enzyme concentrations and increased uptake on gallium scanning can also help in making the diagnosis.³⁹ However, angiotensin-converting-enzyme concentrations are unreliable in the diagnosis of children.

Ocular sarcoidosis might present as acute or chronic uveitis and typical signs include acute anterior uveitis, chronic anterior uveitis, intermediate uveitis, multifocal choroiditis, retinal vasculitis, and optic disc swelling.⁴⁰ There could be characteristic but non-diagnostic signs of ocular sarcoidosis such as mutton-fat keratic precipitates, iris nodules, sheathing along inflamed retinal vessels resembling candle wax drippings, and vitreous opacities resembling strings of pearls. Since ocular sarcoidosis can mimic most other inflammatory ocular diseases, the condition should always be considered in the differential diagnosis of ocular inflammation.

Patients presenting with ocular sarcoidosis might have diagnosable signs of systemic involvement at presentation. However, in those who present with classic ocular signs without evidence of disease in other organ systems, there are no predictors for which patients will go

on to develop systemic sarcoidosis nor pathognomonic ocular signs to diagnose sarcoidosis. Up to 15% of patients will subsequently develop neurological or respiratory involvement and about 45% lose vision from the consequences of ocular sarcoidosis.⁴¹

Uveitis associated with sarcoidosis should be treated on its merits and management depends on the threat to vision. About 50% of patients need systemic steroids or other immunosuppressive medication, such as methotrexate, ciclosporin, and mycophenolate mofetil to control ocular inflammation.^{39,41}

Vogt-Koyanagi-Harada disease

Vogt-Koyanagi-Harada disease is a chronic, bilateral, granulomatous panuveitis, associated with aseptic meningitis, vitiligo, alopecia, and poliosis.⁴² The exact cause of this condition remains unknown, but cell-mediated autoimmunity directed against melanocytes seems to be a unifying mechanism. Vogt-Koyanagi-Harada disease is one of the most common causes of uveitis in Japan and women are more affected than men. It typically has a prodromal neurological and auditory phase, characterised by headache, fever, hyperacusis, dysacusis, and tinnitus, followed by severe panuveitis several days later. Skin and hair changes are late features that are not seen for several months after onset in untreated patients.

Uveitis is a feature of Vogt-Koyanagi-Harada disease and is an acute onset panuveitis characterised by multifocal choroiditis, serous retinal detachments, and pink, swollen, optic discs.⁴³ Uveitis usually leads to the diagnosis being considered, is a serious threat to vision, and represents the major morbidity of the disease. There are several less common patterns of uveitis that could be present in affected patients.

Treatment is high-dose corticosteroid therapy with initial oral doses of 1.5 mg/kg. Vogt-Koyanagi-Harada disease is very sensitive to steroids, with rapid improvement of uveitis and other clinical features on treatment. A slow, tapering course of oral steroids over an extended period is often needed to control the disease and hence minimise the risk of relapse. Patients usually need 12 months or more of corticosteroid therapy and some might need steroid-sparing drugs.

The eye and systemic immunosuppressive treatment

Inflammatory eye disease and its treatment commonly lead to ocular complications. Cataract and secondary glaucoma are the most frequent complications and result from severe or chronic ocular inflammation and local steroid treatment. These are treated surgically, and although the perioperative management can be complex, the outcomes are usually satisfactory with improved vision in most patients.⁴⁴ Macular oedema is another common cause of visual loss, is the major inflammatory-mediated consequence of intraocular

inflammation, and is a principal determinant of anti-inflammatory treatment.

Drugs such as hydroxychloroquine and chloroquine can damage vision because of their toxic effects, which are due to slow accumulation of the drugs in the retinal pigment epithelium that results in irreversible visual loss. Chloroquine is rarely used now that the safer derivative hydroxychloroquine is available. Much debate and confusion have taken place over the type and frequency of ocular examination in patients taking these drugs. The American Academy of Ophthalmology have published evidence-based recommendations for the monitoring of patients on these drugs.⁴⁵

Ocular complications related to systemic immunosuppressive treatment are not uncommon. Patients with severe life-threatening systemic vasculitis who are on high-dose treatment and become septic might develop metastatic fungal or bacterial endophthalmitis from distant septic sites, such as indwelling intravenous catheters, urinary tract infection, and pneumonia. Herpetic retinitis could also develop, especially in those on long-term therapy, and is most frequent in those immunosuppressed after organ transplantation. Opportunistic intraocular infection is rare in patients on immunosuppressive treatment for mainly ocular indications.

Despite improved understanding of the clinical features of inflammatory eye disease and advances in diagnostic testing, clinicians should maintain a high index of suspicion for infective diseases in patients thought to have inflammatory eye involvement. Syphilis has always been the great mimicker of other diseases, and syphilis and other infections (such as Lyme disease) should always be considered in differential diagnosis.

Eye involvement is common in patients with systemic inflammatory disorders. Hence, close collaboration between ophthalmologists and managing physicians is necessary to achieve the best outcome for these patients. Furthermore, initial ocular manifestations might lead to diagnosis of systemic disorders or the eye could become the major focus of the vasculitis resulting in visual loss. Early and optimum immunosuppressive treatment used is essential and new biological drugs (such as anti-TNF α) hold promise in many systemic inflammatory disorders.

Conflict of interest statement

We declare that we have no conflict of interest.

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